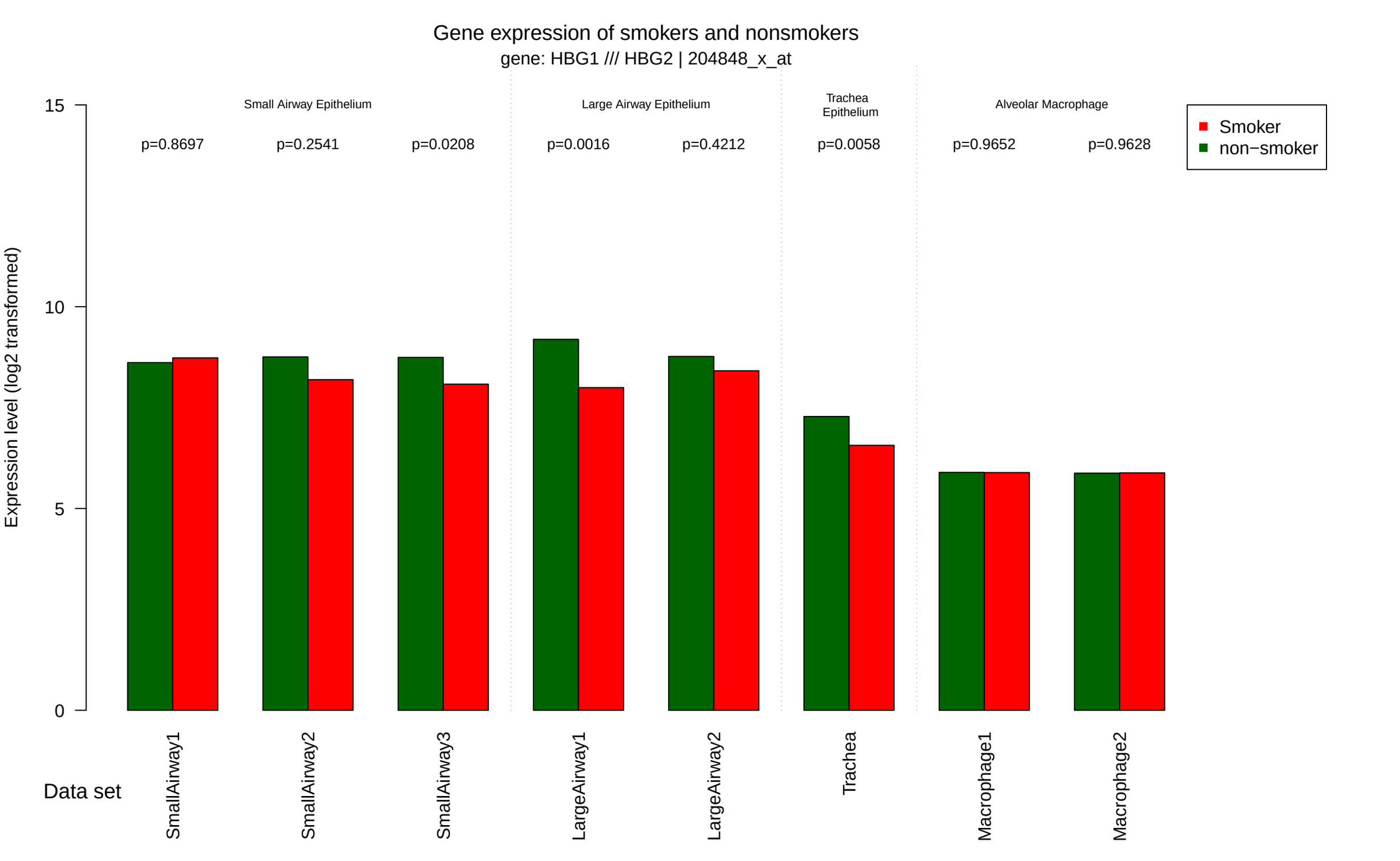
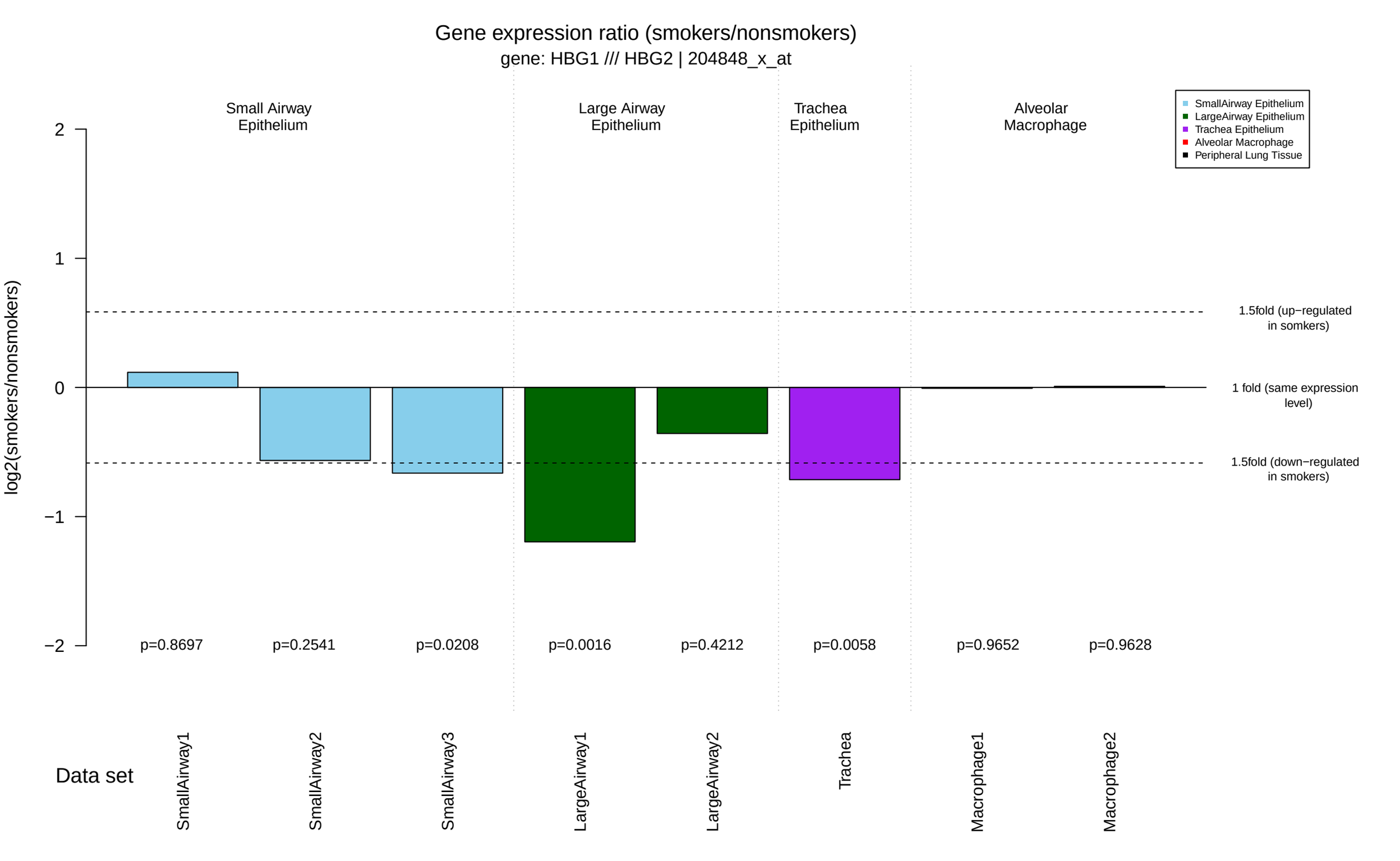
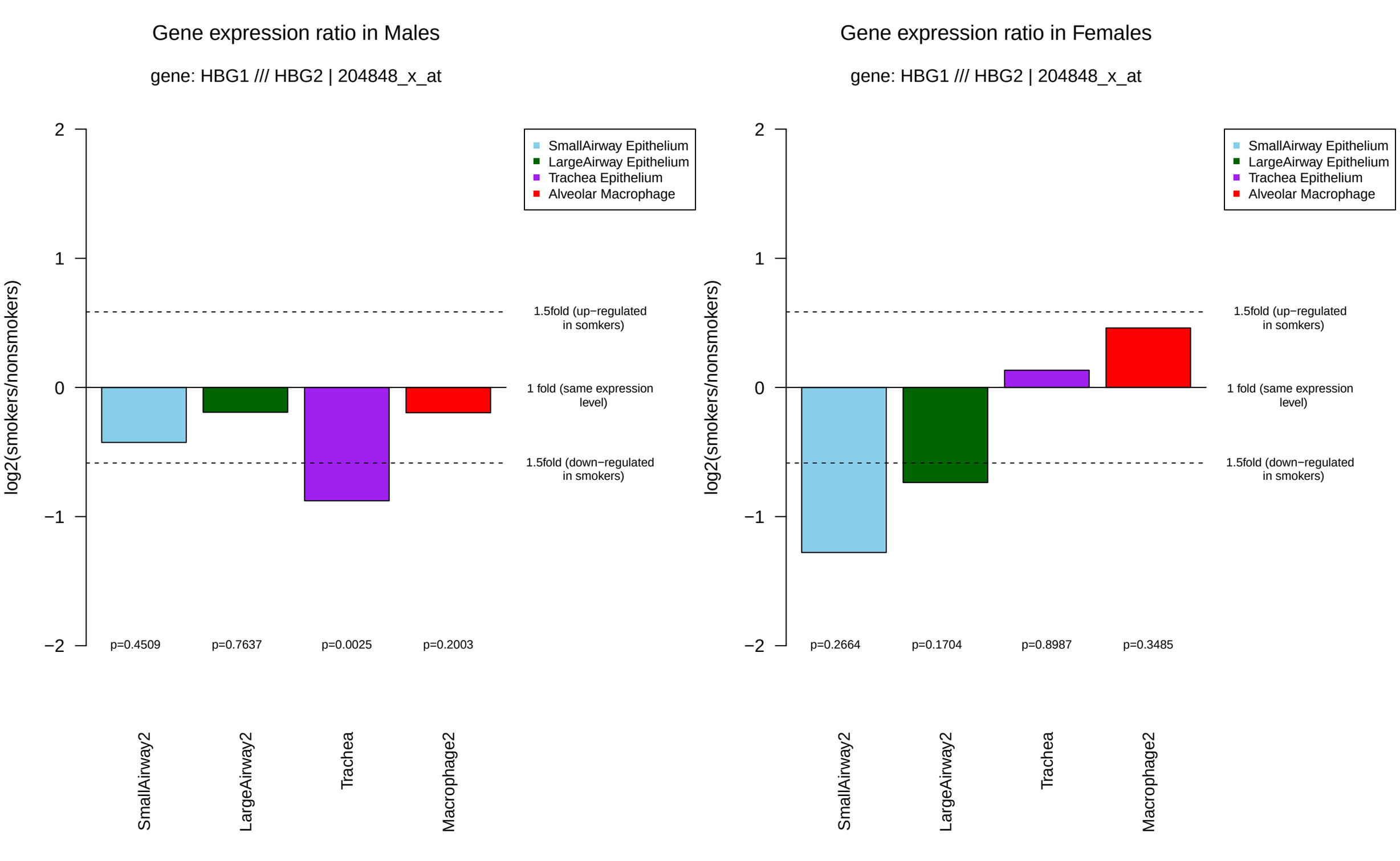
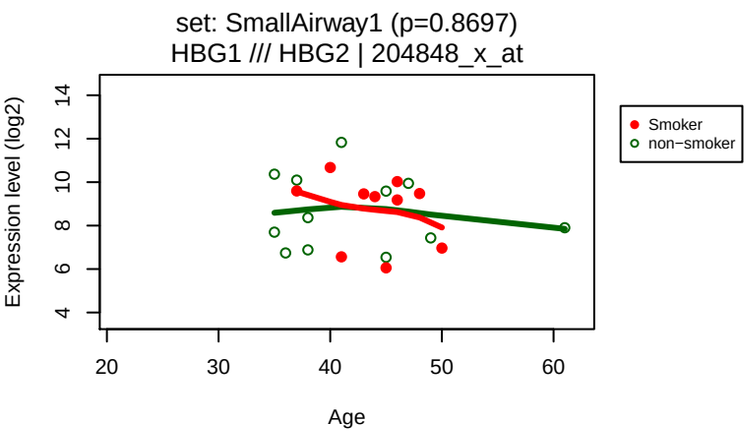
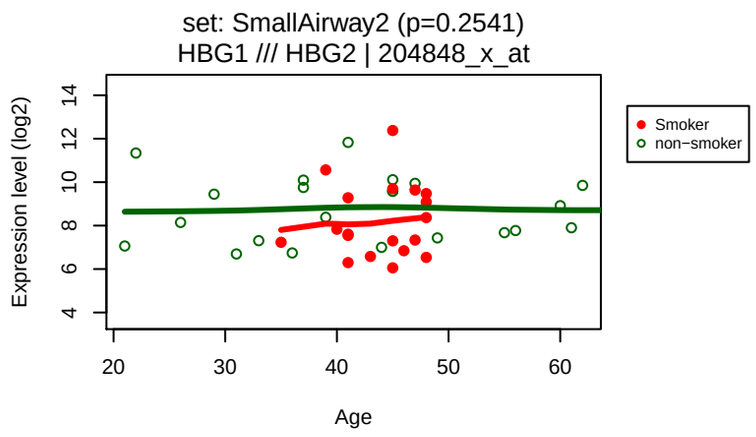
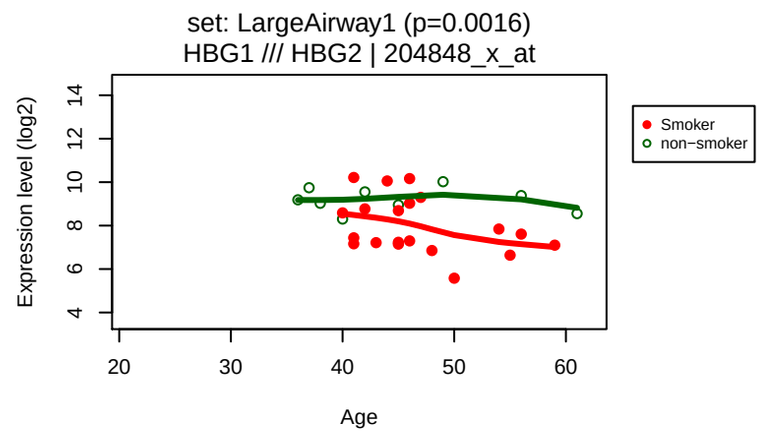

Gene expression of smokers and nonsmokers
gene: HBG1 /// HBG2 | 204848_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8697
p=0.2541
p=0.0208
p=0.0016
p=0.4212
p=0.0058
p=0.9652
p=0.9628
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

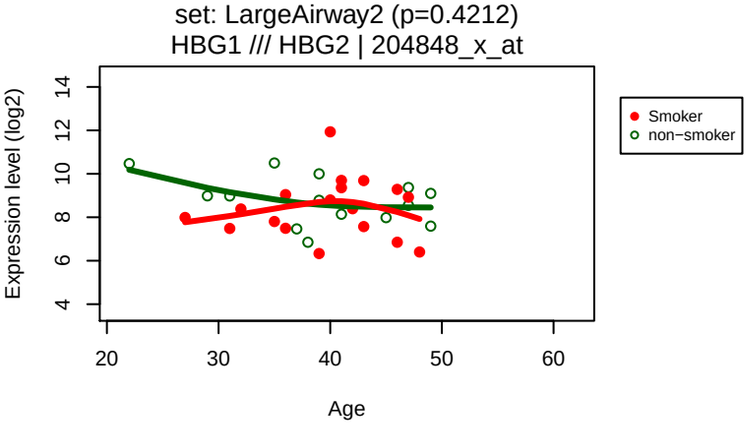

Gene expression ratio (smokers/nonsmokers)
gene: HBG1 /// HBG2 | 204848_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8697
p=0.2541
p=0.0208
p=0.0016
p=0.4212
p=0.0058
p=0.9652
p=0.9628
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

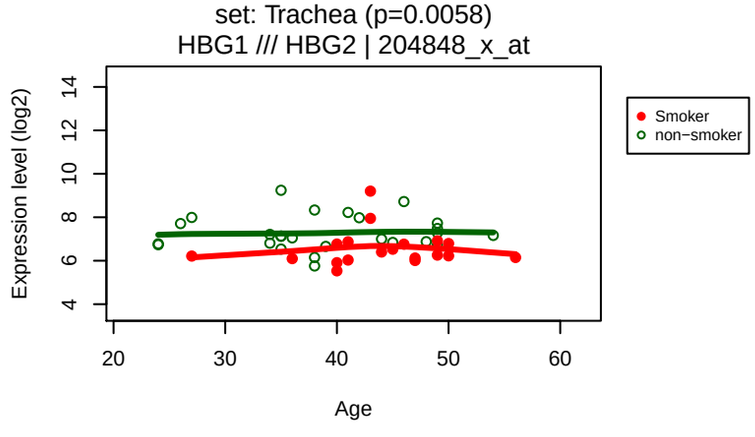

Gene expression ratio in Males
gene: HBG1 /// HBG2 | 204848_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4509
p=0.7637
p=0.0025
p=0.2003
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HBG1 /// HBG2 | 204848_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2664
p=0.1704
p=0.8987
p=0.3485
SmallAirway2
LargeAirway2
Trachea
Macrophage2

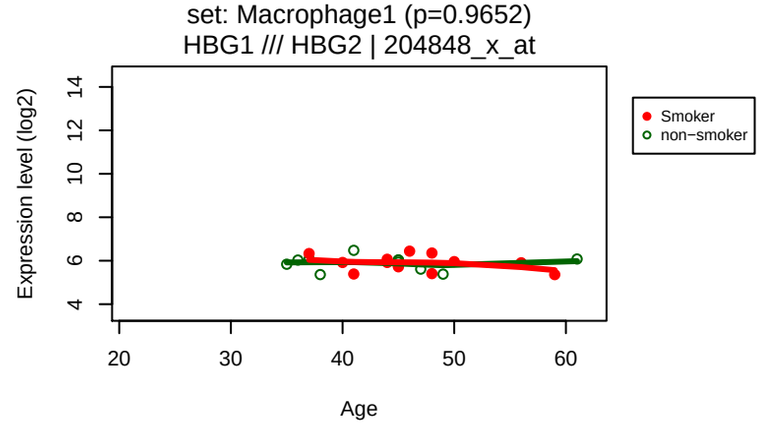

set: SmallAirway1 (p=0.8697)
HBG1 /// HBG2 | 204848_x_at
Expression level (log2)
Age
Smoker
non−smoker

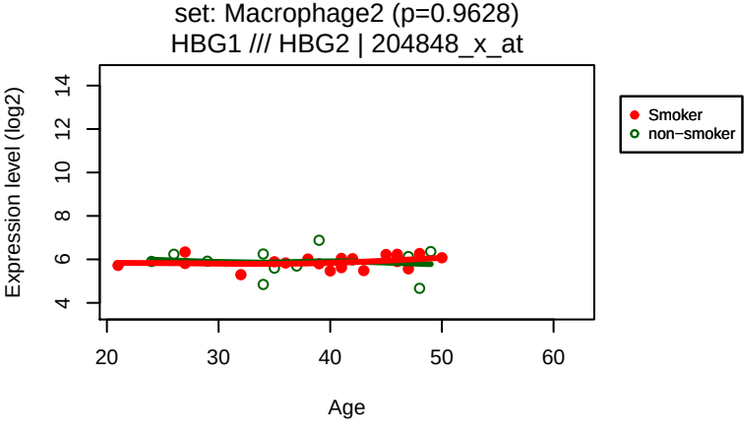

set: SmallAirway2 (p=0.2541)
HBG1 /// HBG2 | 204848_x_at
Expression level (log2)
Age
Smoker
non−smoker

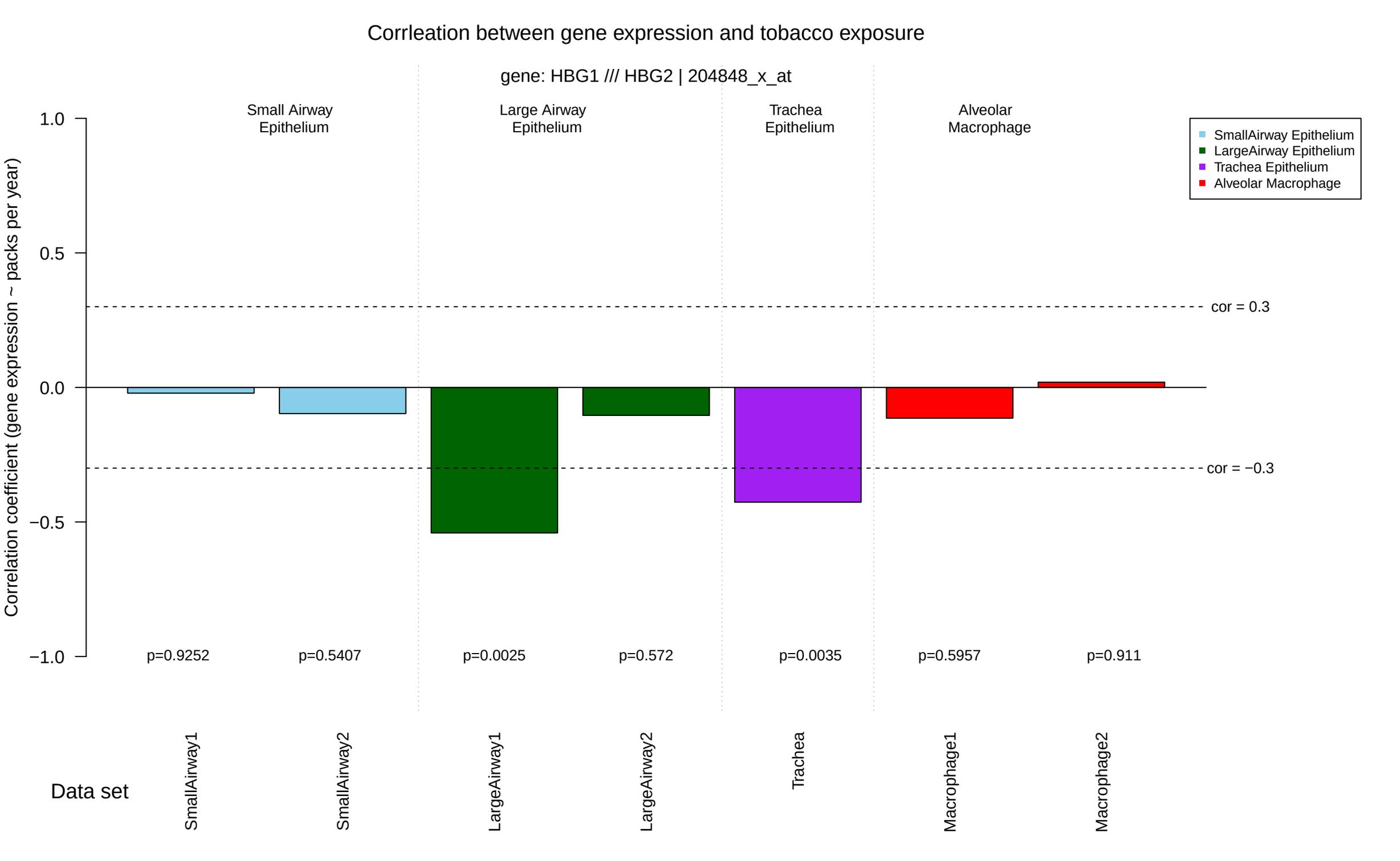

set: LargeAirway1 (p=0.0016)
HBG1 /// HBG2 | 204848_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4212)
HBG1 /// HBG2 | 204848_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0058)
HBG1 /// HBG2 | 204848_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9652)
HBG1 /// HBG2 | 204848_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9628)
HBG1 /// HBG2 | 204848_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HBG1 /// HBG2 | 204848_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9252
p=0.5407
p=0.0025
p=0.572
p=0.0035
p=0.5957
p=0.911
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2